



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Indian Institute of Technology Hyderabad

Assignment-5

ELECTRICAL DEVICES AND CIRCUITS LAB - EE2301

SEMI-III-2024

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Done by
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Objective:

The main objective is to build a 5-bit ADC circuit that measures the input voltage with respect to reference voltage in binary. **Components used:**

1. IC 7474-6
2. IC LM358-1
3. 2k resistors-6
4. 1k resistors-4
5. Arduino
6. LEDs-5

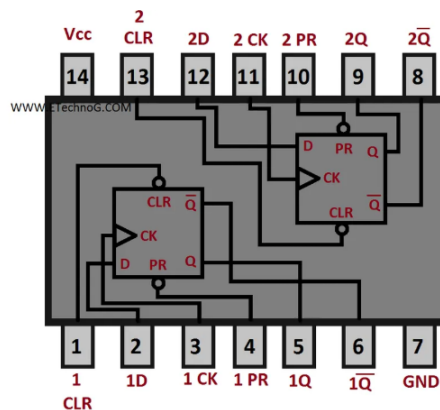


Figure 1: Pin diagram of 7474

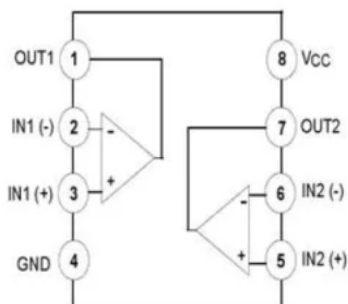


Figure 2: Pin diagram of LM358

Procedure:

The block diagram for this 5-bit ADC is attached below:

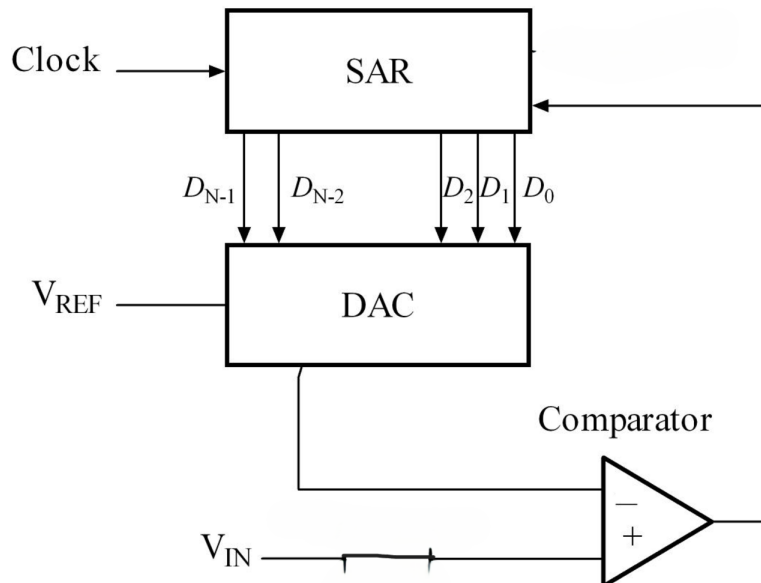


Figure 3: Block diagram (N=5)

The circuit of the SAR block is attached below:

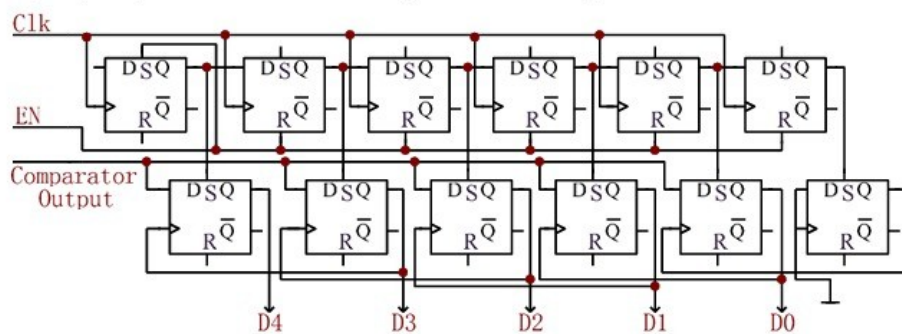


Figure 4: SAR circuit

The circuit diagram of DAC block is attached below:

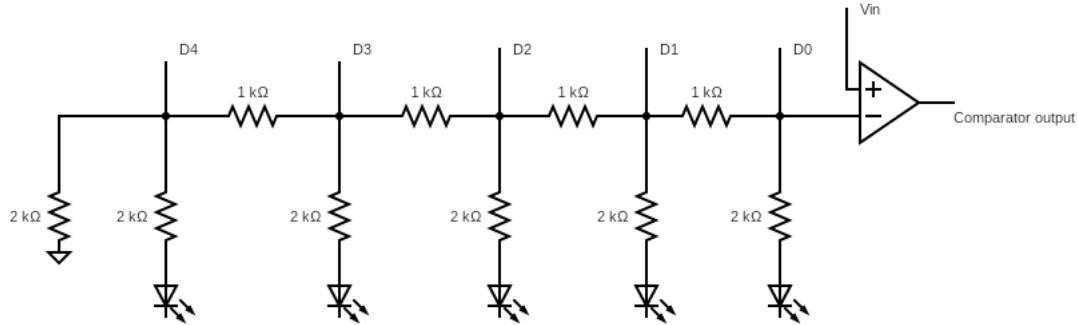


Figure 5: DAC circuit

This ADC circuit is used for displaying the input voltage (V_{in}) in binary with respect to V_{REF} . The value of V_{REF} we used is in the circuit is 2V. Consider V_{in} as the input voltage. With respect to V_{REF} this will be $\frac{31 \times V_{in}}{V_{REF}}$. This value will be represented in binary as the output of the circuit.

The successive approximation register initially sets the most significant bit (MSB) to 1 and all other bits to 0. The DAC output corresponding to this code is compared to the input voltage. If the input voltage is greater, the MSB is kept as it is, and the next bit is set to 1. Otherwise, the MSB is set to 0, and the next bit is set to 1. This process continues for each bit, with the DAC output being compared to the input voltage and the register bits being adjusted accordingly. After five iterations for a five-bit ADC, the final digital code representing the input voltage is obtained. This process can be represented in the picture attached below.

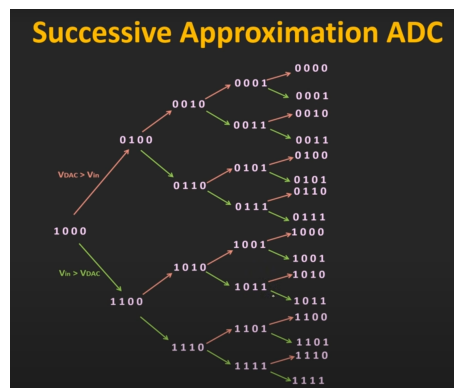


Figure 6

Conclusion:

This way we have successfully constructed an ADC circuit which accurately measures the input voltage and represents in binary form. Some images of our circuit and its output are attached below.

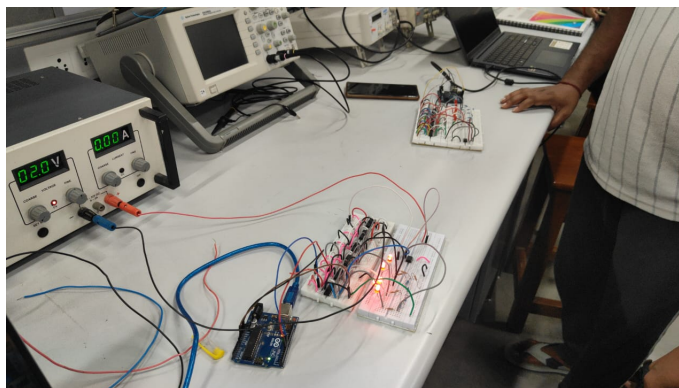


Figure 7

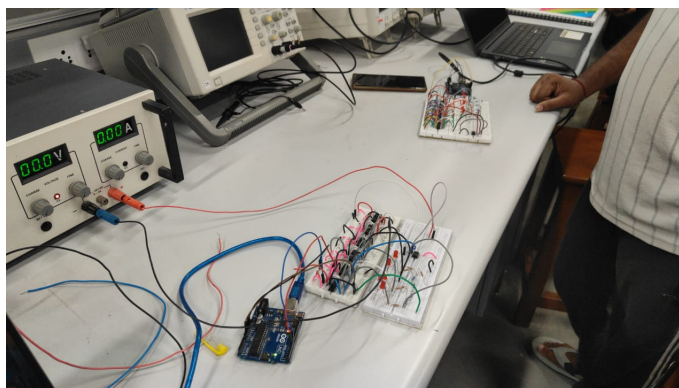


Figure 8

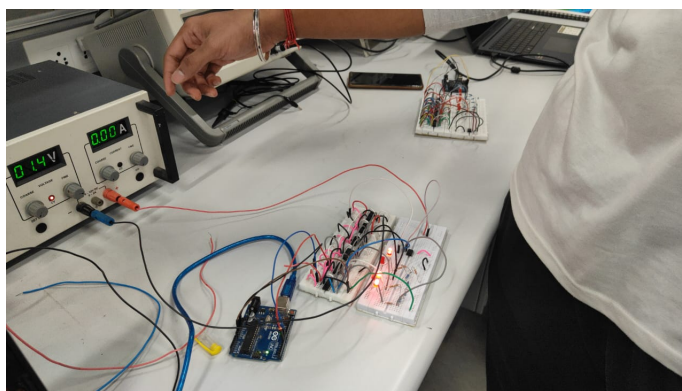


Figure 9